

# Estimation of the Magnitude-Squared Coherence Function Via Overlapped Fast Fourier Transform Processing

G. CLIFFORD CARTER, CHARLES H. KNAPP, and ALBERT H. NUTTALL

**Abstract**—A method for estimating the magnitude-squared coherence function for two zero-mean wide-sense-stationary random processes is presented. The estimation technique utilizes the weighted overlapped segmentation fast Fourier transform approach. Analytical and empirical results for statistics of the estimator are presented. The analytical expressions are limited to the nonoverlapped case; empirical results show a decrease in bias and variance of the estimator with increasing overlap and suggest a 50-percent overlap as being highly desirable when cosine (Hanning) weighting is used.

## 1. Introduction

The complex coherence function is the normalized cross spectral density,

$$\gamma_{xy}(f) = \frac{\Phi_{xy}(f)}{\sqrt{\Phi_{xx}(f)\Phi_{yy}(f)}} \quad (1)$$

where  $\Phi_{xy}(f)$  is the cross spectral density at frequency  $f$  between two zero-mean stationary random processes  $x(t)$  and  $y(t)$  with auto spectra  $\Phi_{xx}(f)$  and  $\Phi_{yy}(f)$ . In particular, the magnitude-squared coherence (MSC)

$$|\gamma_{xy}(f)|^2 = \frac{|\Phi_{xy}(f)|^2}{\Phi_{xx}(f)\Phi_{yy}(f)} \quad (2)$$

has useful properties, which has led investigators to examine the errors arising from estimating this function [1]–[18].

Current techniques for estimating the MSC generally use the fast Fourier transform (FFT) [19]–[21]. Briefly, the method consists of obtaining two finite-time series from the random processes being investigated. Each time series is partitioned into  $n$  equal length segments and sampled at  $P$  equally

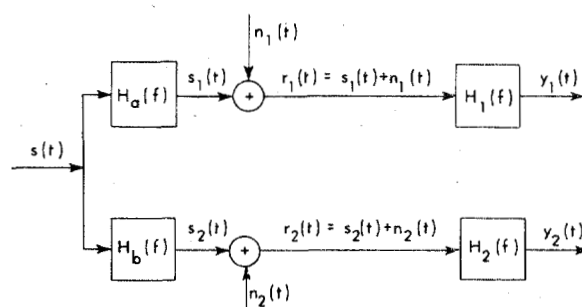


Fig. 1. Configuration for examining properties of MSC.

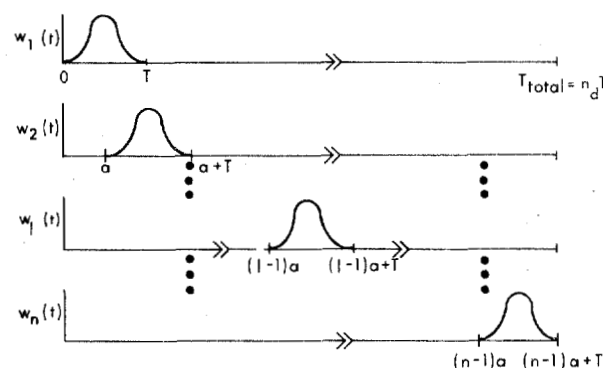


Fig. 2. Overlapped weighting functions.

spaced data points. The segments may be overlapped or disjoint. Samples from each segment are multiplied by a weighting function, and the FFT of the weighted  $P$ -point sequence is performed. Then the Fourier coefficients for each weighted segment are used to estimate the auto- and cross-power spectral densities. The spectral density estimates thus obtained are used in (2) to form the MSC estimate.

Spectral resolution of the estimates varies inversely with the segment length  $T$ . Proper weighting or “windowing” of the  $T$ -seconds segment is also helpful in achieving good sidelobe reduction. On the other hand, for independent segments with ideal windowing, the bias and the variance of the MSC estimate vary inversely with the number of segments  $n$ . Therefore, to generate a good estimate with limited data, one may be faced with conflicting requirements on  $n$  and  $T$ . Segment overlapping can be used to increase both  $n$  and  $T$ . As the percentage of overlap increases, however, the computational requirements increase rapidly, while the improvement stabilizes owing to the greater correlation between data segments [16].

This paper presents the results of two studies that provide helpful insight into the choice of  $n$  and the amount of overlap for MSC estimation for a given frequency resolution  $T^{-1}$ . In the first study, results of Fisher [1] and Goodman [2] were utilized to obtain analytical expressions for the probability distribution and density functions of the MSC estimate and, thence, for the bias and variance of the estimate.

Manuscript received July 18, 1972.

G. C. Carter and A. H. Nuttall are with the Naval Underwater Systems Center, New London Laboratory, New London, Conn. 06320.

C. H. Knapp is with the Department of Electrical Engineering, University of Connecticut, Storrs, Conn. 06268.

The results are valid for Gaussian processes with independent data segments and ideal windowing.

In the second study, the effect of overlap on MSC estimation is examined through a computational study that uses cosine (Hanning) weighting, a finite time history, and a specified frequency resolution. Under these conditions, estimates of the bias and variance of the MSC estimate have been calculated as a function of overlap for two values of coherence.

Results of these studies are preceded by brief discussions of the properties of the MSC and its estimation.

## II. Properties of the MSC Function

In order to determine the basic properties and certain applications of the MSC function, special cases of the configuration depicted in Fig. 1 will be considered. It is assumed that  $s(t)$ ,  $n_1(t)$ , and  $n_2(t)$  in Fig. 1 are sample functions of stationary uncorrelated random processes. By the application of basic relationships, it may be verified that [18]

$$|\gamma_{y_1 y_2}(f)|^2 = |\gamma_{r_1 r_2}(f)|^2 \quad (3)$$

$$= \frac{\Phi_{ss}^2(f) |H_a(f) H_b^*(f)|^2}{[\Phi_{ss}(f) |H_a(f)|^2 + \Phi_{n_1 n_1}(f)] [\Phi_{ss}(f) |H_b(f)|^2 + \Phi_{n_2 n_2}(f)]} \quad (4)$$

Examination of (2) and (4) and Fig. 1 yields the following conclusions.

1) The MSC is a real number, conveniently normalized to lie between zero and unity.

2) Since  $|\gamma_{y_1 y_2}(f)|^2$  is independent of  $H_1(f)$  and  $H_2(f)$  in (4), the MSC between two signals is not affected by the arbitrary linear processing of these signals.

3) With  $n_1(t) = n_2(t) = 0$  in Fig. 1,  $|\gamma_{y_1 y_2}(f)|^2 = 1$ . Consequently, the MSC of two signals derived from linear operations on the same signal is unity. Equivalently, the MSC of two linearly related signals is unity.

4) If  $\Phi_{n_1 n_1}(f) = \Phi_{n_2 n_2}(f) = \Phi_{nn}(f)$  and  $|H_a(f)|^2 = |H_b(f)|^2 = 1$ , then

$$|\gamma_{y_1 y_2}(f)|^2 = \frac{\Phi_{ss}^2(f)}{[\Phi_{ss}(f) + \Phi_{nn}(f)]^2} \quad (5)$$

and the signal-to-noise power ratio (S/N) is

$$\frac{\Phi_{ss}(f)}{\Phi_{nn}(f)} = \frac{|\gamma_{y_1 y_2}(f)|}{1 - |\gamma_{y_1 y_2}(f)|} \quad (6)$$

5) If  $n_2(t) = 0$  and  $|H_a(f)|^2 = 1$ , then

$$|\gamma_{y_1 y_2}(f)|^2 = \frac{\Phi_{ss}^2(f)}{[\Phi_{ss}(f) + \Phi_{nn}(f)] \Phi_{ss}(f)} \quad (7)$$

and the S/N is

$$\frac{\Phi_{ss}(f)}{\Phi_{nn}(f)} = \frac{|\gamma_{y_1 y_2}(f)|^2}{1 - |\gamma_{y_1 y_2}(f)|^2} \quad (8)$$

Conclusions 4) and 5) indicate ways in which the MSC can be used to estimate the S/N. Conclusion 3) suggests that the MSC can be used as a measure of linearity.

## III. Estimation of the MSC

We assume throughout the paper that  $x(t)$  and  $y(t)$  are sample functions of random processes that are zero-mean, wide-sense stationary, and ergodic. The basic MSC estimation procedure is to obtain estimates of the spectral densities  $\Phi_{xx}(f)$ ,  $\Phi_{xy}(f)$ ,  $\Phi_{yy}(f)$  and use these estimates in (2) in place of their exact counterparts. The method to be described for generating power spectral density estimates is the direct method discussed in [8]–[18]. It employs (cosine) weighting, overlapped processing, and the FFT.

Let the time-limited sample functions  $x(t)$  and  $y(t)$  be sampled (at a frequency  $f_s$  Hz that is greater than twice the larger bandwidth of the two processes) and converted to two (quantized) sequences of numbers or time series. Each time series is effectively divided

into  $n$  segments through the multiplication of a series of real weighting functions, as illustrated in Fig. 2. If  $T$  is the length of each segment,  $n_d$  is the possible number of disjoint segments, and  $n_d T$  seconds is the total record length, then the total number of (overlapped) segments is

$$n = \frac{n_d - O_f}{1 - O_f} \quad (9)$$

where

$$O_f = \frac{T - a}{T} = 1 - \frac{a}{T} \quad (10)$$

is the overlap fraction. When  $O_f$  is zero,  $n = n_d$ .

A  $P$ -point ( $P = f_s T$ ) FFT is obtained for the  $P$ -point sequence representing each of the weighted data segments. The frequency domain equivalent of multiplying each segment by a weighting function is a convolution of the spectrum with the Fourier transform of the weighting function. Ideally, therefore, the weighting function transform should be an impulse, which would require a unit weighting for all  $t$ . Equivalently,  $T$  should be large. Practical factors affecting selection of the segment length,  $T$  and window shape include the following:

1)  $n_d = T_{\text{total}}/T$  should be large in order to reduce the bias and variance of the spectral estimates.

2) The Fourier transform of the weighting function  $w_l(t)$  should have a main lobe that is narrower than

TABLE I  
Probability Density and Distribution Functions

| Density Function                                                                                                                                                                                                                        |      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| $p( \hat{\gamma} ^2   n_d,  \gamma ^2) = (n_d - 1) (1 -  \gamma ^2)^{n_d} (1 -  \hat{\gamma} ^2)^{n_d-2}$                                                                                                                               |      |
| $\cdot {}_2F_1(n_d, n_d; 1;  \gamma ^2  \hat{\gamma} ^2), \quad 0 \leq  \gamma ^2  \hat{\gamma} ^2 < 1$                                                                                                                                 | (Ia) |
| $= (n_d - 1) (1 -  \gamma ^2)^{n_d} (1 -  \hat{\gamma} ^2)^{n_d-2}$                                                                                                                                                                     |      |
| $\cdot (1 -  \gamma ^2  \hat{\gamma} ^2)^{1-2n_d} {}_2F_1(1 - n_d, 1 - n_d; 1;  \gamma ^2  \hat{\gamma} ^2)$                                                                                                                            | (Ib) |
| $= (n_d - 1) \left[ \frac{(1 -  \gamma ^2)(1 -  \hat{\gamma} ^2)}{(1 -  \gamma ^2  \hat{\gamma} ^2)^2} \right]^{n_d}$                                                                                                                   |      |
| $\cdot \frac{(1 -  \gamma ^2  \hat{\gamma} ^2)}{(1 -  \hat{\gamma} ^2)^2} {}_2F_1(1 - n_d, 1 - n_d; 1;  \gamma ^2  \hat{\gamma} ^2).$                                                                                                   | (Ic) |
| Distribution Function                                                                                                                                                                                                                   |      |
| $P( \hat{\gamma} ^2   n_d,  \gamma ^2) =  \hat{\gamma} ^2 \left( \frac{1 -  \gamma ^2}{1 -  \gamma ^2  \hat{\gamma} ^2} \right)^{n_d} \sum_{k=0}^{n_d-2} \left( \frac{1 -  \hat{\gamma} ^2}{1 -  \gamma ^2  \hat{\gamma} ^2} \right)^k$ |      |
| $\cdot {}_2F_1(-k, 1 - n_d; 1;  \gamma ^2  \hat{\gamma} ^2).$                                                                                                                                                                           | (Id) |

the finest detail of the true spectra of processes  $x(t)$  and  $y(t)$ . This lobe width is inversely proportional to  $T$ .

3) In order to minimize the sidelobes of the Fourier transform of  $w_l(t)$ , higher order derivatives of  $w_l(t)$  should be continuous, if possible.

Thus good spectral resolution (large  $T$ ) and small bias and variance (large  $n$ ) provide conflicting requirements on  $n$  and  $T$  for a specified record length  $T_{\text{total}}$ .

If  $X_l(f_k)$  denotes the FFT of the  $l$ th weighted segment of  $x(t)$  at frequency  $f_k$ , then the spectral density estimates are given by

$$\begin{aligned}\hat{\Phi}_{xx}(f_k) &= \alpha \sum_{l=1}^n |X_l(f_k)|^2, \\ \hat{\Phi}_{xy}(f_k) &= \alpha \sum_{l=1}^n X_l(f_k) Y_l^*(f_k), \\ \hat{\Phi}_{yy}(f_k) &= \alpha \sum_{l=1}^n |Y_l(f_k)|^2,\end{aligned}\quad (11)$$

where  $\alpha = 1/(nPf_s)$ . Finally, the MSC estimate is

$$|\hat{\gamma}_{xy}(f_k)|^2 \triangleq \frac{|\hat{\Phi}_{xy}(f_k)|^2}{\hat{\Phi}_{xx}(f_k) \hat{\Phi}_{yy}(f_k)}. \quad (12)$$

Observe that when  $n = 1$ ,  $|\hat{\gamma}_{xy}(f_k)|^2 = 1$ , regardless of the true value of MSC. Consequently, the estimate is biased, in general, the actual bias being a function of  $n$  and true MSC,  $|\gamma_{xy}(f)|^2$ .

Note that if  $O_f = 0$ , there will be no overlap, and each segment is virtually independent of the previous one (except for correlated edge effects). All theoretical results presented in the next section are valid for the case of independent segments, that is, no

overlap. A detailed analysis of the effect of overlapped weighted segmentation for estimating auto spectral densities is given in [16].

#### IV. Statistics for the Estimate of the MSC

Goodman [2] derived an analytical expression for the probability density function of the magnitude-coherence estimate  $|\hat{\gamma}|$  based on (12).<sup>1</sup> His results are valid for two Gaussian random processes that have been partitioned into  $n_d$  independent segments. Each segment was assumed large enough to ensure adequate spectral resolution. Furthermore, each segment was assumed perfectly weighted in the sense that the Fourier coefficient at the  $k$ th frequency received no power from other bins. (However, Hannan [13] points out that the statistics do not hold at the zeroeth or folding frequencies.)

The first-order probability density and distribution functions for the estimate of MSC, given the true value of MSC,  $|\gamma|^2$ , and the number of independent segments  $n_d$  are given in Table I. Equation (Ia) in Table I is obtained directly by using the relation

$$p(|\hat{\gamma}|^2 | n_d, |\gamma|^2) = \frac{1}{2|\hat{\gamma}|} p(|\hat{\gamma}| | n_d, |\gamma|^2). \quad (13)$$

Alternate expressions (Ib) and (Ic) are obtained with the help of [22, eq. (15.35)]. Equations (Ib) and (Ic) are useful because  ${}_2F_1(1 - n_d, 1 - n_d; 1; |\gamma|^2 |\hat{\gamma}|^2)$  is an  $(n_d - 1)$ st-order polynomial in  $|\gamma|^2 |\hat{\gamma}|^2$ .

Fisher [1], working on statistics of the estimate of the squared correlation coefficient, derived the probability density for that random variable. He inte-

<sup>1</sup>The  $f$  dependence and subscripts are suppressed for notational simplicity.

grated the result and achieved a closed-form solution for the distribution function; specifically, the solution was a finite sum of  ${}_2F_1$  functions, each one a polynomial. Although these statistics are for a different problem, proper identification of the variables yields exactly the integration formula needed to find the probability distribution of the estimate of MSC, namely, (Id) in Table I.

Figs. 3 and 4 illustrate the probability density and distribution functions for several cases, as computed using (Ib) and (Id) from Table I. It is evident from Fig. 4 that the bias and variance of the MSC estimate decrease when  $n_d$  is increased.

The bias and variance of the MSC estimate can be evaluated using a general expression for the  $m$ th moment of the MSC estimate. This expression, obtained from (Ia) in Table I and [23, eq. (7.512 12)], is

$$\begin{aligned} E[(|\hat{\gamma}|^2)^m | n_d, |\gamma|^2] &= \int_{-\infty}^{+\infty} p(|\hat{\gamma}|^2 | n_d, |\gamma|^2) (|\hat{\gamma}|^2)^m d|\hat{\gamma}|^2 \\ &= (1 - |\gamma|^2)^{n_d} \frac{\Gamma(n_d) \Gamma(m+1)}{\Gamma(n_d + m)} \\ &\quad \cdot {}_3F_2(m+1, n_d, n_d; m+n_d, 1; |\gamma|^2) \end{aligned} \quad (14)$$

where

$${}_3F_2(a, b, c; d, e; z) = \sum_{k=0}^{\infty} \frac{(a)_k (b)_k (c)_k}{(d)_k (e)_k} \frac{z^k}{k!} \quad (15)$$

$$(a)_k \triangleq \frac{\Gamma(a+k)}{\Gamma(a)}. \quad (16)$$

These results can be verified through the proper identification of the variables in the work of Anderson [3], who extended Fisher's original work on the squared correlation coefficient [1].

Bias and variance expressions obtained using (14) are summarized in Table II. Approximations (IIc) and (IId) are the result of truncating the series (IIa) and (IIb). Equations (IIe) through (IIg) then follow for large  $n_d$ ; they indicate that the MSC estimate is asymptotically unbiased, and that for large  $n_d$  the following is true.

1) The bias is greatest,  $1/n_d$ , at  $|\gamma|^2 = 0$ , and smallest, 0, at  $|\gamma|^2 = 1$ .

2) The variance is zero for  $|\gamma|^2 = 1$  and greatest,  $(2/3)^3/n_d$ , at  $|\gamma|^2 = 1/3$ .

3) The mean-square error from the true value is equal to the variance, provided  $|\gamma|^2 \neq 0$ .

Equation (IIe) is a more accurate result than the empirically determined bias,  $(1 - |\gamma|^2)/n_d$ , given in [12]. However, Benignus' result, (IIIf), is an upper bound on our approximation (IIe).

Figs. 5 and 6, respectively, show the bias and variance as functions of  $n_d$  and  $|\gamma|^2$ . For values of

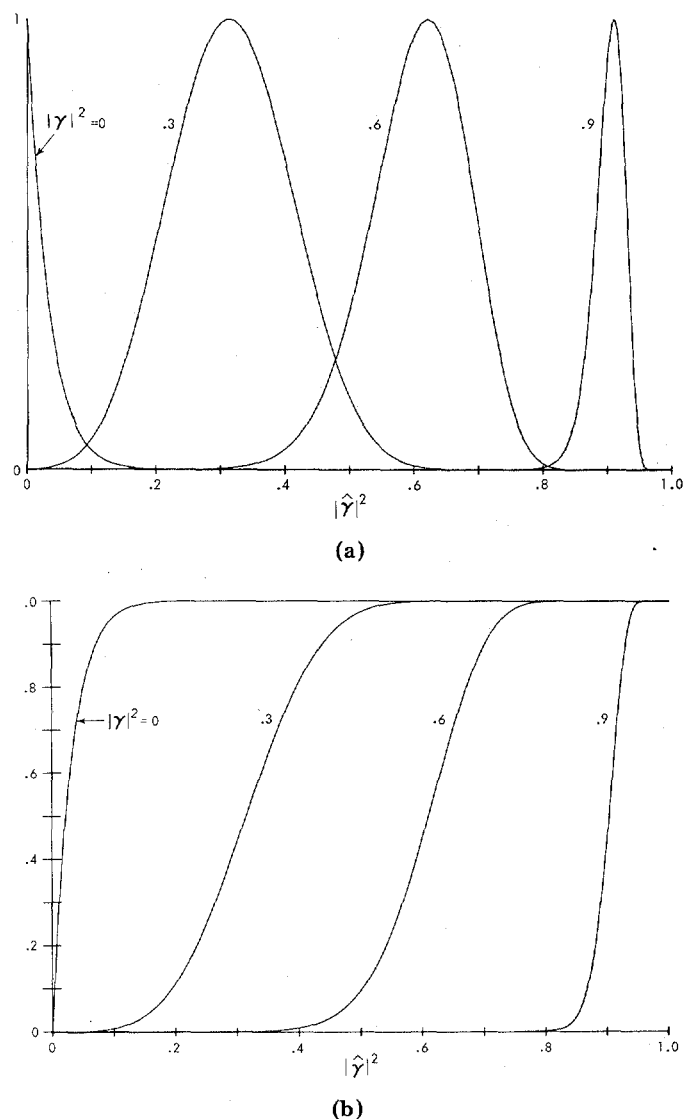


Fig. 3. (a) Probability density functions. (Functions have been normalized by maximum values, which are 31.0, 4.13, 5.23, and 17.5.) (b) Distribution functions.

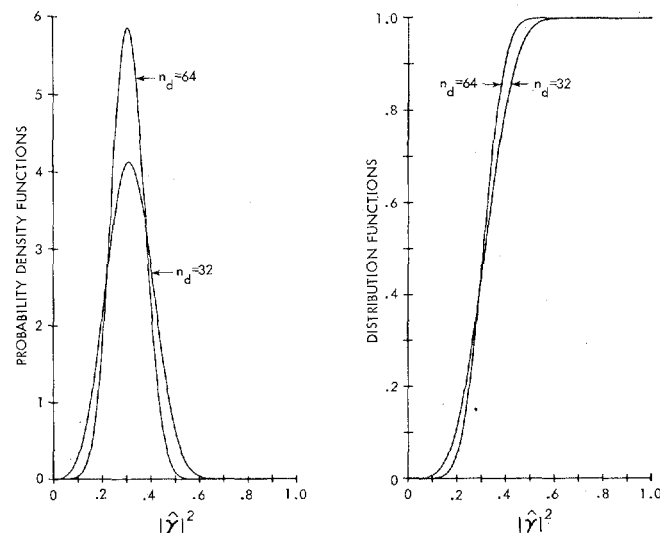


Fig. 4. Probability density and distribution functions of  $|\hat{\gamma}|^2$  for  $|\gamma|^2 = 0.3$ .

TABLE II  
Bias and Variance of the Estimation Error

| Exact                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bias                          | $B = E[\hat{\gamma}^2   n_d, \gamma^2] - \gamma^2 = \frac{(1 - \gamma^2)^{n_d}}{n_d} {}_3F_2(2, n_d, n_d; n_d + 1, 1; \gamma^2) - \gamma^2$ $= \frac{1}{n_d} + \frac{n_d - 1}{n_d + 1} \gamma^2 {}_2F_1(1, 1; n_d + 2; \gamma^2) - \gamma^2. \quad (\text{IIa})$                                                                                                                                                                                                                                                                           |
| Variance                      | $V = E(\hat{\gamma}^4   n_d, \gamma^2) - E^2(\hat{\gamma}^2   n_d, \gamma^2)$ $= \frac{2(1 - \gamma^2)^{n_d}}{n_d(n_d + 1)} {}_3F_2(3, n_d, n_d; n_d + 2, 1; \gamma^2)$ $- \left[ \frac{(1 - \gamma^2)^{n_d}}{n_d} {}_3F_2(2, n_d, n_d; n_d + 1, 1; \gamma^2) \right]^2. \quad (\text{IIb})$                                                                                                                                                                                                                                               |
| Approximate                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| $B_0$                         | $B_0 \cong \frac{1}{n_d} - \frac{2}{n_d + 1} \gamma^2 + \frac{1!(n_d - 1)}{(n_d + 1)(n_d + 2)} (\gamma^2)^2 + \frac{(n_d - 1)2!}{(n_d + 1)(n_d + 2)(n_d + 3)} (\gamma^2)^3; \quad B \cong \begin{cases} B_0, & B_0 \geq 0 \\ 0, & B_0 < 0 \end{cases} \quad (\text{IIc})$                                                                                                                                                                                                                                                                  |
| $V_0$                         | $V_0 \cong \frac{(n_d - 1)}{n_d(n_d + 1)} \left[ \frac{1}{n_d} + 2 \frac{n_d - 2}{n_d + 2} \gamma^2 - 2 \frac{2n_d^3 - n_d^2 - 2n_d + 3}{(n_d + 1)(n_d + 2)(n_d + 3)} (\gamma^2)^2 \right.$ $+ 2 \frac{n_d^4 - 6n_d^3 - n_d^2 + 10n_d - 8}{(n_d + 1)(n_d + 2)(n_d + 3)(n_d + 4)} (\gamma^2)^3$ $\left. + \frac{13n_d^5 - 15n_d^4 - 113n_d^3 + 27n_d^2 + 136n_d - 120}{(n_d + 1)(n_d + 2)^2(n_d + 3)(n_d + 4)(n_d + 5)} (\gamma^2)^4 \right]; \quad V \cong \begin{cases} V_0, & V_0 \geq 0 \\ 0, & V_0 < 0 \end{cases} \quad (\text{IId})$ |
| Approximation for Large $n_d$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| $B$                           | $B \cong \frac{1}{n_d} [1 - \gamma^2]^2 \quad (\text{IIe})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $B$                           | $\leq \frac{1}{n_d} [1 - \gamma^2] \quad (\text{IIF})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| $V$                           | $V \cong \begin{cases} \frac{1}{n_d^2}, & \gamma^2 = 0 \\ \frac{2\gamma^2}{n_d} (1 - \gamma^2)^2, & 0 < \gamma^2 \leq 1 \end{cases} \quad (\text{IIg})$                                                                                                                                                                                                                                                                                                                                                                                    |

$n_d$  in the range from 32 to 64, expressions (IIe) through (IIg) are good approximations; however, the curves in Figs. 5 and 6 were obtained using the exact formulas, (IIa) and (IIb). Peaks in the variance curves at  $\gamma^2 \cong 1/3$  are evident.

## V. Experimental Investigation of Overlap Effects

An experimental study has been made of the effect of overlap of data on the MSC estimate,  $\hat{\gamma}^2$ . The analytical results presented earlier relate only to the case of independent segments, that is, the case of zero overlap. Intuitively, the application of non-overlapped weighting functions does not make the most efficient use of the data when forming the estimate,  $\hat{\gamma}^2$ . The experiment described herein examines this inefficiency, in terms of bias and variance of  $\hat{\gamma}^2$ , as a function of different amounts of overlap.

The method of evaluating overlap is straightforward in concept. Data are generated with an accurately prespecified value of MSC that is independent of frequency. Then the sample mean and variance of  $\hat{\gamma}^2$  can be measured for the given overlap by averaging over frequency. In order to generate two processes with an MSC independent of frequency, let

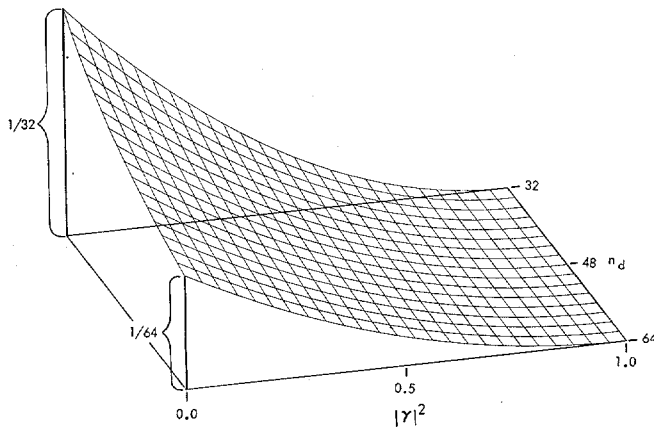
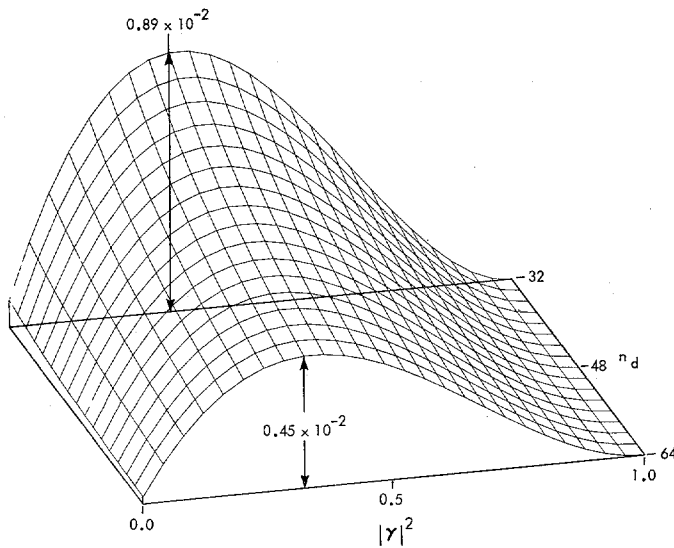
$$\begin{aligned} x(t) &= n_1(t) + Gn_2(t) \\ y(t) &= n_2(t) + Gn_1(t) \end{aligned} \quad (17)$$

where

$$\Phi_{n_1 n_2}(f) = 0, \Phi_{n_1 n_1}(f) = \Phi_{n_2 n_2}(f) = \Phi_{nn}(f).$$

It follows from (2) and (3) that (using more explicit notation again)

$$|\gamma_{xy}(f)|^2 = \frac{4G^2 \Phi_{nn}^2(f)}{(1 + G^2)^2 \Phi_{nn}^2(f)} = \frac{4G^2}{(1 + G^2)^2} \quad (18)$$

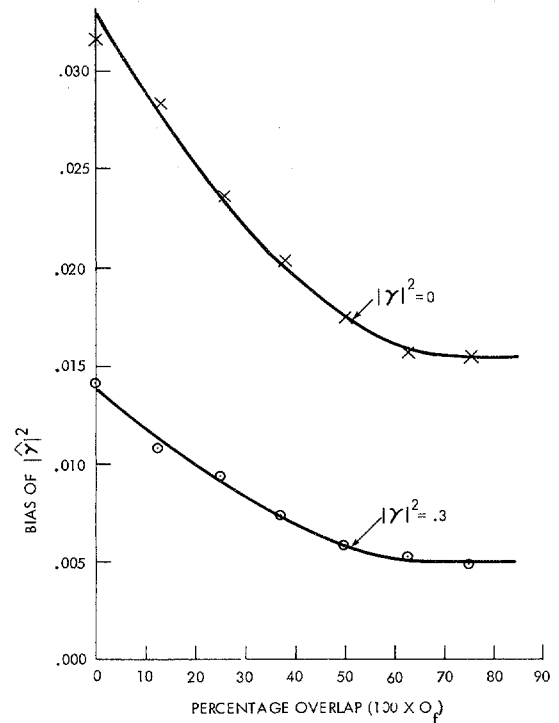
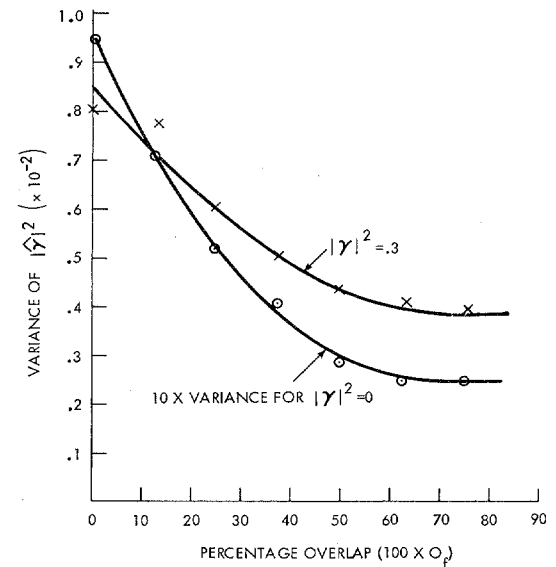
Fig. 5. Bias of  $|\hat{\gamma}|^2$  versus  $|\gamma|^2$  and  $n_d$ .Fig. 6. Variance of  $|\hat{\gamma}|^2$  versus  $|\gamma|^2$  and  $n_d$ .

an expression that is independent of frequency. The gain  $G$  can be selected according to (18) to achieve a desired value of  $|\gamma_{xy}(f)|^2$ . The processes generated according to (17) can be shown to be maximally insensitive to minor differences in the power spectral density functions of the original uncorrelated waveforms,  $n_1(t)$  and  $n_2(t)$ .

In order to generate variable-coherence time series, one Gaussian noise source was used to generate a time-limited (32-s) sample function of  $n_1(t)$  and, later, of  $n_2(t)$ . (This method eliminates the need for two identical filters.) The waveforms were first band limited to 1200 Hz using a low-pass filter and then digitized. Next, the digital data were stored on magnetic tape in a format compatible with overlapped processing. Digital versions of  $x(t)$  and  $y(t)$  were generated from digital versions of  $n_1(t)$  and  $n_2(t)$  for two values of  $|\gamma_{xy}(f)|^2$ .

Pertinent information regarding the data includes

- 1) bandwidth of  $\Phi_{nn}(f) = 1200$  Hz
- 2) record length of  $x(t)$  and  $y(t) = 32$  s

Fig. 7. Bias of  $|\hat{\gamma}|^2$  when  $n_d = 32$ .Fig. 8. Variance of  $|\hat{\gamma}|^2$  when  $n_d = 32$ .

- 3) sampling rate  $f_s = 4096$  Hz
- 4) FFT interval  $T = 1$  s (4096 samples).

Estimates of the bias and variance were performed according to

$$\hat{\text{bias}} = \left[ \frac{1}{1000} \sum_{k=1}^{1000} |\hat{\gamma}_{xy}(f_k)|^2 \right] - |\gamma_{xy}(f)|^2 \quad (19)$$

and

$$\hat{\text{var}} = \frac{1}{999} \sum_{k=1}^{1000} [|\hat{\gamma}_{xy}(f_k)|^2 - \hat{\text{bias}}]^2 \quad (20)$$

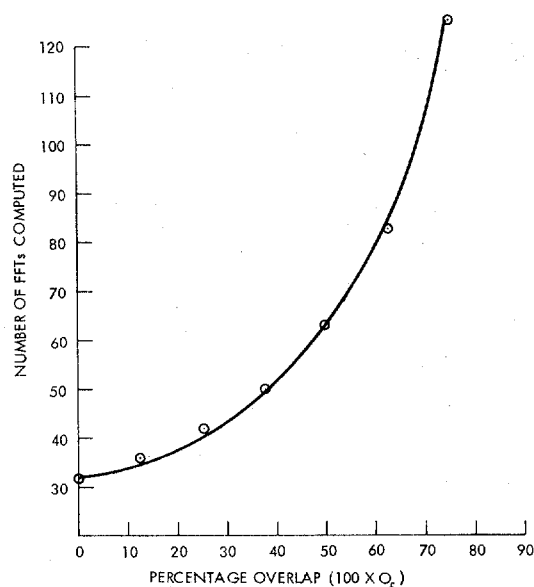


Fig. 9. Number of FFT's required for overlapped processing when  $n_d = 32$ .

Results of the experiment are summarized in Figs. 7 and 8. It is apparent from these results that the bias and variance of  $|\hat{\gamma}_{xy}(f_k)|^2$  can be reduced through overlapped processing. For example, when  $|\gamma_{xy}(f_k)|^2 = 0.0$ , the variance of the estimator with a 50-percent overlap equals 31 percent of the variance of the estimator with no overlap. With a 50-percent overlap, the bias is 55 percent as large as with no overlap. Similarly, when  $|\gamma_{xy}(f)|^2 = 0.3$  and the overlap is 50 percent, the variance is 55 percent of the nonoverlapped estimator, and the bias is 50 percent as large. Observe that the bias and variance for zero overlap agree very well with (Ile) through (IIg) in Table II. With a 62.5-percent overlap, or greater, the bias and variance achieve values corresponding to an effective  $n_d$  of about 64 with nonoverlapped processing.

Clearly, as the overlap increases, the computational cost must also increase. Fig. 9 [a plot of (9) with  $n_d = 32$ ] illustrates the number of FFT's per sample function required as a function of overlap. Increasing the overlap from 50 to 62.5 percent requires 32 percent more FFT's, but the variance of the MSC estimator decreases only from 80 to 95 percent of its value at a 50-percent overlap. It is doubtful, therefore, that the improvement derived from using a 62.5-percent overlap, as opposed to a 50-percent overlap, will warrant the increased computational costs, except in unusual circumstances. Overlap percentages of 25 to 50 percent are quite reasonable, however.

## VI. Conclusions

A detailed analytical analysis of the statistics for estimating the MSC function has been made. When such estimates are made, time-limited sample func-

tions of long duration, which are stationary (in the wide sense) over the period of observation, must be available. Expressions for the probability density, the cumulative distribution, and the bias and variance of the MSC estimate have been presented for Gaussian processes and for the case where nonoverlapped processing is used. Evaluation of these expressions, which are dependent on both the true value of the MSC and the number of disjoint segments  $n_d$ , dramatically portrays the requirement that  $n_d$  be large.

The application of a cosine-weighting function, in order to reduce errors from sidelobe leakage, makes inefficient use of the available data. As shown empirically, much better use of these data can be achieved through overlapped processing. An overlap of about 62.5 percent provides the limit of improvement achievable. This limit is roughly equivalent to having twice as much data as with nonoverlapped processing. The reduced bias and variance of the estimator achieved through 62.5-percent overlapped processing can be realized almost entirely through a 50-percent overlap. The computational cost associated with 50-percent overlap is not unreasonable.

## Acknowledgment

The authors wish to acknowledge the assistance of Dr. H. M. Lucal and Dr. R. J. Kochenburger of the University of Connecticut, for reviewing portions of the manuscript; Dr. P. G. Cable and Dr. F. J. Kingsbury of the Naval Underwater Systems Center, for their support; R. M. Kennedy, R. R. Kneipfer, C. R. Arnold, and J. F. Ferrie, of the Naval Underwater Systems Center, for their recommendations; and J. C. Sikorski of the Naval Underwater Systems Center, for digitizing the data.

## References

- [1] R. A. Fisher, *Contributions to Mathematical Statistics*. New York: Wiley, 1950. (Chapter 14 was originally published as "The general sampling distribution of the multiple correlation coefficient," in *Proc. Roy. Soc.*, ser. A, vol. 121, pp. 654-673, 1928.)
- [2] N. R. Goodman, "On the joint estimation of the spectra, cospectrum, and quadrature spectrum of a two-dimensional stationary Gaussian process," New York Univ., New York, Sci. Paper 10 (AD 134 919), Mar. 1957.
- [3] T. W. Anderson, *An Introduction to Multivariate Statistical Analysis*. New York: Wiley, 1958.
- [4] D. E. Amos and L. H. Koopmans, "Tables of the distribution of the coefficient of coherence for stationary bivariate Gaussian processes," Sandia Corp., Mono. SCR-483, 1963.
- [5] R. A. Haubrich, "Earth noise, 5 to 500 millicycles per second. I. Spectral stationarity, normality, and non-linearity," *J. Geophys. Res.*, vol. 70, pp. 1415-1427, Mar. 1965.
- [6] L. D. Enochson and N. R. Goodman, "Gaussian approximations to the distribution of sample coherence," Air Force Flight Dynamics Lab., Res. Tech. Div., AF Systems Command, Wright-Patterson AFB, Ohio, Bull. AFFDL-TR-65-67, June 1965.
- [7] L. J. Tick, "Estimation of Coherency," in *Spectral Analysis of Time Series*, B. Harris, Ed. New York: Wiley, pp. 133-152, 1967.
- [8] C. Bingham, M. D. Godfrey, and J. W. Tukey, "Modern

- techniques of power spectrum estimation," *IEEE Trans. Audio Electroacoust.*, vol. AU-15, pp. 56-66, June 1967.
- [9] P. D. Welch, "The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms," *IEEE Trans. Audio Electroacoust.*, vol. AU-15, pp. 70-73, June 1967.
- [10] G. M. Jenkins and D. G. Watts, *Spectral Analysis and Its Applications*. San Francisco, Calif.: Holden-Day, 1968.
- [11] C. H. Knapp, "An algorithm for estimation of the inverse spectral matrix," General Dynamics Electric Boat Res. Proj., SUBIC Rep. 1970-U417-70-010, Apr. 1969.
- [12] V. A. Benignus, "Estimation of the coherence spectrum and its confidence interval using the fast Fourier transform," *IEEE Trans. Audio Electroacoust.*, vol. AU-17, pp. 145-150, June 1969.
- [13] E. S. Hannan, *Multiple Time Series*. New York: Wiley, 1970.
- [14] P. R. Roth, "Effective measurements using digital signal analysis," *IEEE Spectrum*, vol. 8, pp. 62-70, Apr. 1971.
- [15] G. C. Carter and A. H. Nuttall, "Statistics of the estimate of coherence," *Proc. IEEE (Lett.)*, vol. 60, pp. 465-466, Apr. 1972.
- [16] A. H. Nuttall, "Spectral estimation by means of overlapped FFT processing of windowed data," Naval Underwater Systems Cent., New London, Conn., Rep. 4169, Oct. 1971.
- [17] J. S. Bendat and A. G. Piersol, *Random Data: Analysis and Measurement Procedures*. New York: Wiley, 1971.
- [18] G. C. Carter, "Estimation of the magnitude-squared coherence function (spectrum)," Naval Underwater Systems Cent., New London, Conn., Tech. Rep. TR 4343 (AD 743 945), May 19, 1972.
- [19] J. W. Cooley and J. W. Tukey, "An algorithm for the machine calculation of complex Fourier series," *Math. Comput.*, vol. 19, pp. 297-301, Apr. 1965.
- [20] R. C. Singleton, "An algorithm for computing the mixed radix fast Fourier transform," *IEEE Trans. Audio Electroacoust.*, vol. AU-17, pp. 93-102, June 1969.
- [21] B. Gold and C. M. Rader, *Digital Processing of Signals*. New York: McGraw-Hill, 1969.
- [22] M. Abramowitz and I. A. Stegun, Eds., *Handbook of Mathematical Functions With Formulas, Graphs, and Mathematical Tables*. Washington, D.C.: U.S. Gov. Printing Office, 1964.
- [23] I. S. Gradshteyn and I. M. Ryzhik, *Table of Integrals, Series, and Products*. New York: Academic Press, 1965.

# A Spectral Characterization of the Ill-Conditioning in Numerical Deconvolution

M. P. EKSTROM

**Abstract**—An important class of applications in digital signal processing involves the numerical solution of the convolution integral. These so-called numerical deconvolution problems are notoriously difficult to solve because of their inherent ill-conditioning. In this paper we present a characterization of this ill-conditioning based on a classical spectral decomposition of the discrete convolution. Factors prominently influencing the conditioning are identified and some explicit sensitivity measures are introduced.

## I. Introduction

The numerical solution of the convolution integral over a finite region is a classical problem occurring in many fields of engineering. Particular examples in digital signal processing are encountered in seismology

[1], pulse communications [2], signal decomposition [3], and image restoration [4]. Convolution-type integral equations are of the form

$$g(t) = \int_0^b h(t-t') f(t') dt' \quad (1)$$

with  $g(t)$  defined for all  $t \in [0, b]$ . In most physical applications the functions in (1) can be taken to be square integrable, differentiable to all finite orders, and of finite bandwidth. With  $g(t)$  and  $h(t)$  known, the problem of determining  $f(t)$  is called deconvolution. The initial step in numerically solving (1) usually involves its discretization. By dividing the interval into  $N$  equally spaced regions and applying a general class of quadrature forms, the integral can be represented by a so-called convolution summation

$$g(t_n) = \sum_{n'=0}^N H(t_n - t_{n'}) f(t_{n'}) \quad (2)$$

where  $T = b/N$  and  $t_n = nT$ . The numerical deconvolution problem consists of solving this linear system.

It is well known that, in its general form, deconvolution is not a well-posed problem. As Phillips [5] pointed out, while the mapping defined by the convolution integral is well behaved (continuous and linear over the space of square integrable functions), its inverse may not exist and, in those cases where it does exist, it is generally unbounded. The manifestation of this property on the discrete representation in (2) is the resulting singularity or, at best, ill-conditioning of the system of equations. By ill-conditioning, we mean that small perturbations in  $g(t_n)$  give rise to

Manuscript received October 16, 1972; revised December 16, 1972. This paper was supported in part by the U.S. Atomic Energy Commission.

The author is with the Lawrence Livermore Laboratory, University of California, Livermore, Calif. 94550.